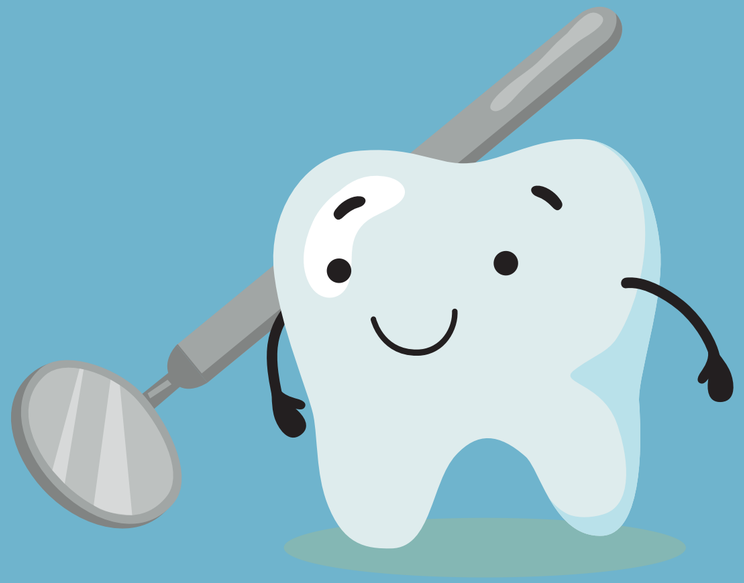




Full Veneer Crown Prep



A FULL GUIDE

WHAT IS A FULL VENEER PREP?

- A full veneer crown prep is:
  a tooth preparation where **ALL** surfaces of the tooth are reduced.
- Allows the placement of a:

FULL COVERAGE CROWN



Preparation Criteria

- Main Goal: Create enough space for the **crown material** while **preserving as much of the healthy tooth structure as possible.**

A good prep should have:

- The correct amount of occlusal and axial reduction.
- Smooth line angles.
- Well defined finish line.
- Proper taper for **retention** and **resistance.**



Measuring the Amount of Reduction

To measure the amount of reduction accurately, use a condensation silicon impression using the technique demonstrated in the video below:



What you need for the Perfect prep

You will need the following instruments:



Micromotor for low speed



Contra low/high speed



Mirror+explorer



Probe



Tapered with round end blue



Needle blue



Flame/Football blue



Flame/Football yellow



Tapered with round end yellow

The Occlusal Surface

The most frequently asked question by students: How much reduction should be done on the occlusal surface?

To answer this question, you need to know a few things first:

- Is the tooth **upper** or **lower**?
- Which cusp is the **Functional cusp? Buccal or Lingual?**
- What will be the material used for the future crown?

The answer to these questions will help you determine the amount of reduction you need.



The Occlusal Surface

Generally speaking, Occlusal reduction for full coverage crowns is **approximately 1.5mm overall.**

- Functional cusp =**1.5-2mm**
- Non-Functional cusp= **1-1.5mm**

Why don't we reduce all the cusps equally?



The answer is simple: not all cusps receive the same occlusal forces.

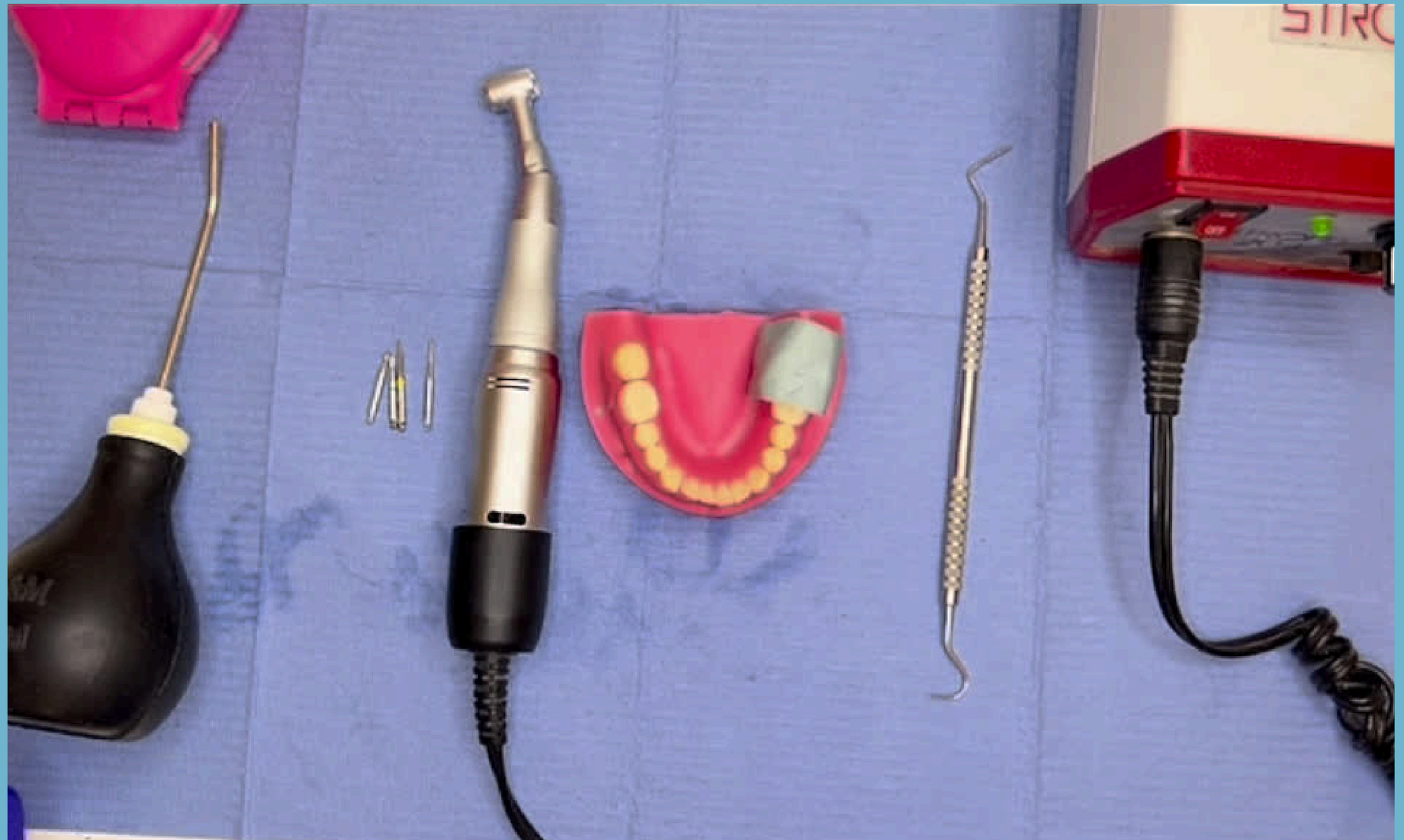
The functional cusp receives **more occlusal load** so it needs **more reduction** to create a **greater thickness of crown material to withstand the forces.**

The non functional cusp receives **less forces so needs less reduction.**



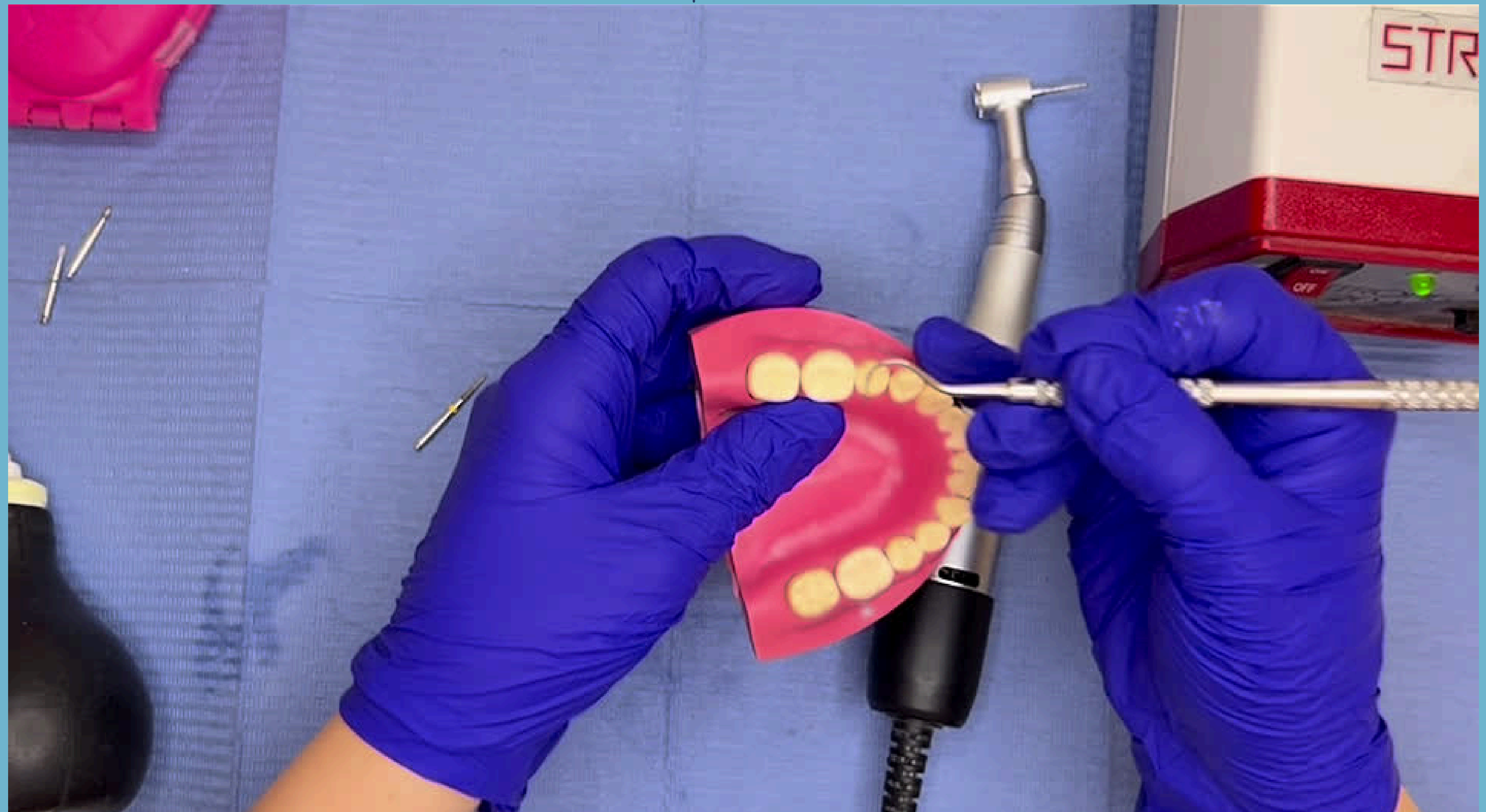
The Occlusal Surface

Let’s take a look at the istruments needed:



The Occlusal Surface

Some important notes :



The Occlusal Surface

Now you know the **WHY**

Let's see the **HOW**



Axial Reduction

Without any fancy or complicated words, Axial Reduction just means the reduction done on any surface of the tooth **except the occlusal surface.**

<u>To master axial reduction, you have to understand three concepts:</u>

- What is the long Axis of the tooth?
- How to position and move your contra parallel to the long axis of the tooth.
- Is your goal to “draw” your finish line? Or does it just appear?



Let’s discuss each one in detail!

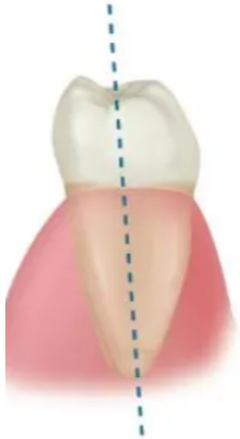
(NOTE: You are a few slides away from being a crown prep **BOSS!**)

Axial Reduction

The Long Axis of the Tooth: easily the most misunderstood imaginary line of all imaginary lines! It sounds complicated but it's just a line passing through the **center** of the tooth.

Notice that it is **NOT** a straight line. It follows the **natural tilt** and **anatomy** of the tooth.





Axial Reduction

In this video, I will show you how to position and move your contra along the long axis of the tooth: easy peasy



Axial Reduction

The fastest way to ruin your crown prep:

Thinking that your goal is to **“draw”** the finish line using your stone.

I have seen students do this a million times and it always ends in the same way: **REMAKE**



To avoid this happening to you, understand that a proper finish line is **NOT A GOAL** but it is a **RESULT.**

This means that:

- If you reduce the proper amount of tooth structure
- If you move your contra **parallel** to the long axis on all surfaces

You will get the perfect finish line without even trying or doing any extra step.

Axial Reduction

Now let's put all that information together and prep axial surfaces of this tooth.



Proximal Reduction

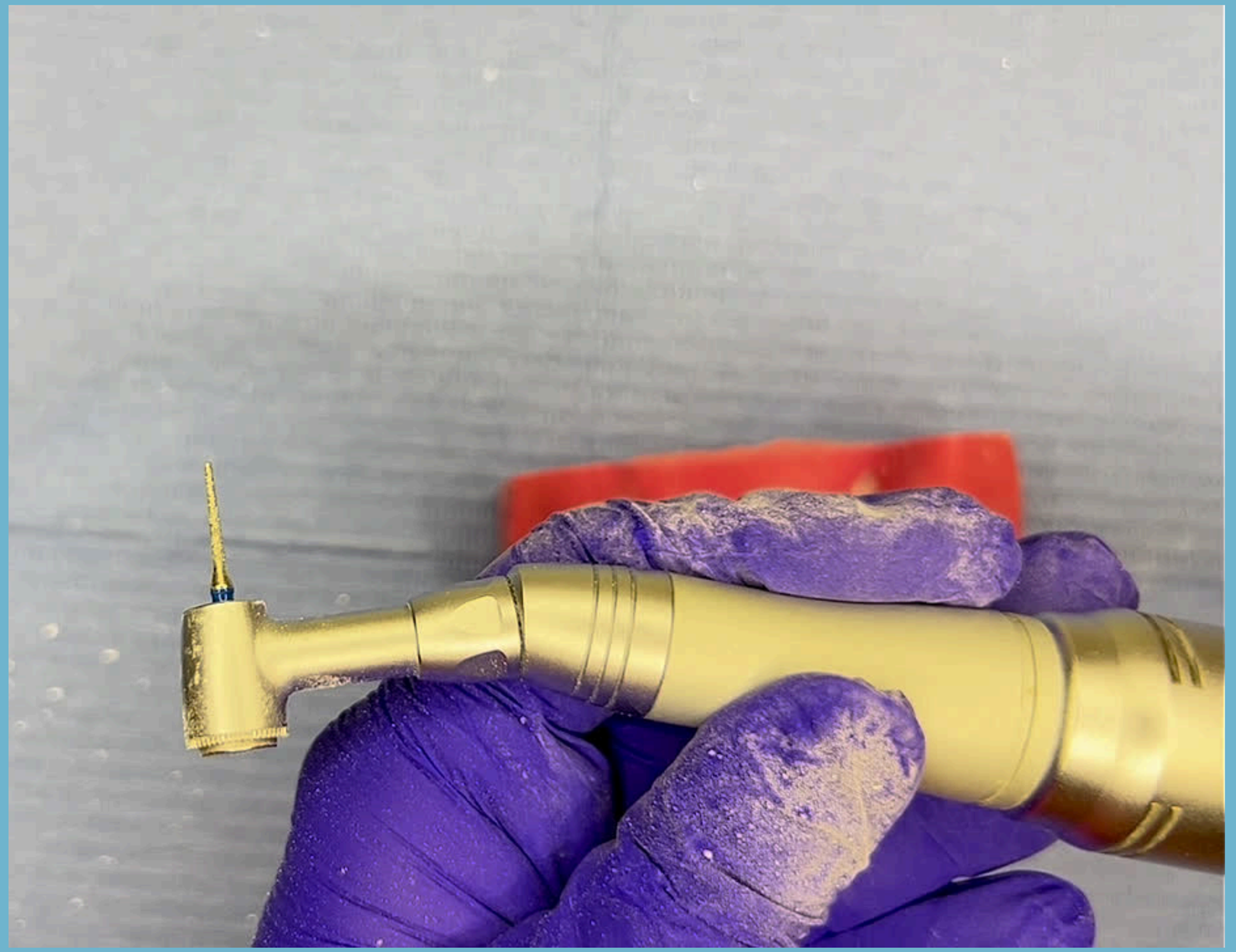
It is part of axial reduction but with few extra things to consider. (sadly)

What you need to know:

- When reducing a proximal surface (Mesial/Distal), you need to use a **Needle Stone** to open the **Contact** with the **adjacent tooth.**
- Keep using the needle stone till there is enough space for Tapered with Round End to pass **smoothly.**
- **NEVER** use a needle stone to create your finish line. The function of needle stone is to open the contact and create space for tapered with round **ONLY!**
- Keep your hand parallel to the long axis **at all times!**



Proximal Reduction



Proximal Reduction

When prepping a proximal surface, the next logical question to ask is :

How to prevent injury to the adjacent tooth?

Injury can be prevented by these methods:

- Creating an **Enamel Lip** to act as a shield for the adjacent tooth. How to do it correctly will be shown in a video.

- Using a matrix band to protect the adjacent tooth while opening the contact. .



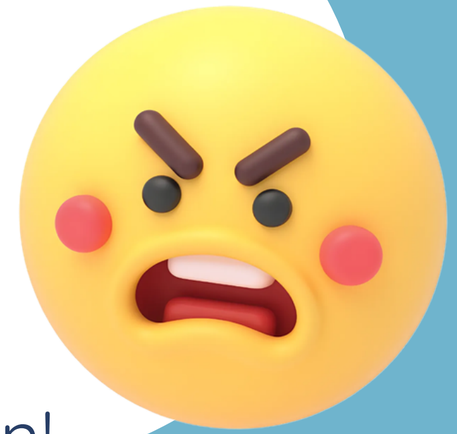
Enamel Lip



Did I open the contact properly?

This question has driven EVERY dental student crazy!

Did you ever show your prep to your instructor and they said: ‘the contact needs to be opened a **bit** more” or maybe they said : “you opened it **too much.”**



And you’re just standing there confused.

Don’t worry! We got you! Never be confused again!

Just watch the video in the next slide.

Checking the Contact



Final Touches

If you made it this far, well done! You are just a step away from being a crown prep master!

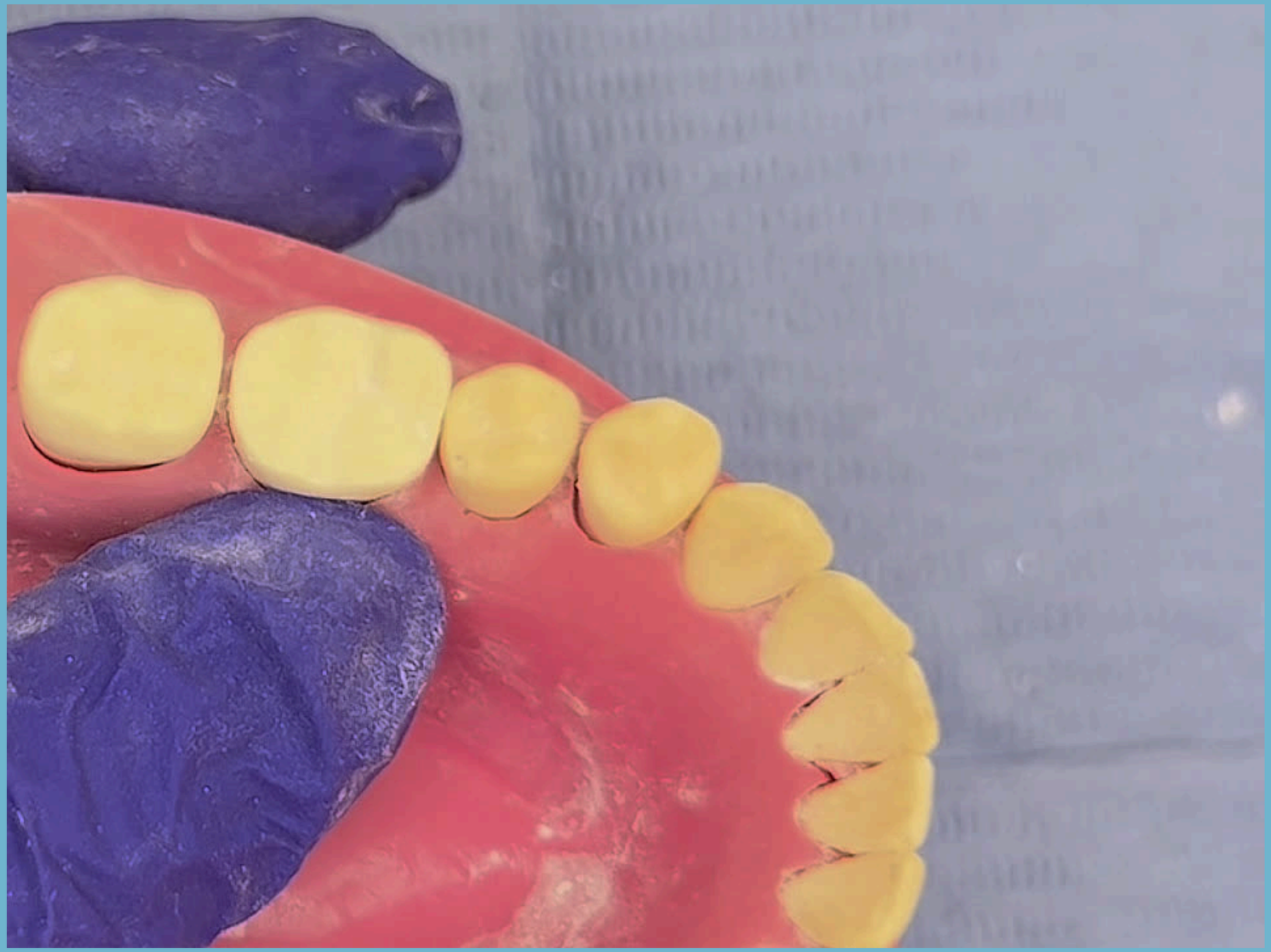


Functinal cusp bevel vs Secondary Plane

- FCB is done by lightly **bevelling/removing** the **cusp tips** of the **functional** cusps.
- Secondary plane is done on the **occlusal third** of the **buccal surface** at a 45 degree angle.

Sounds confusing we know. Just watch the video on the next slide.

Functional cusp bevel



Secondary plane



Finishing

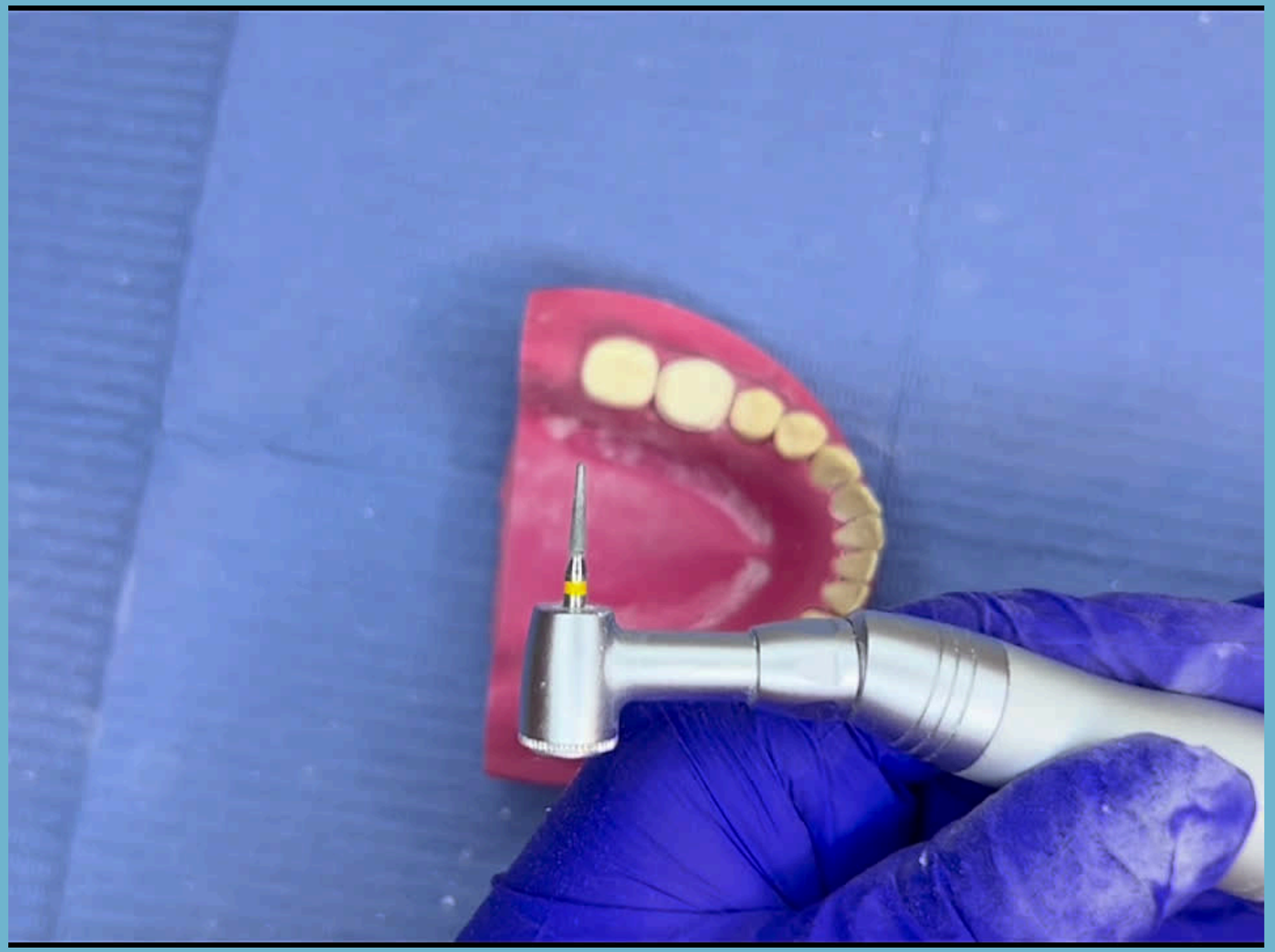


We promise you after this step, your'e really done!
Let's talk about finishing your prep.
This step is often skipped by students thinking that it does not matter. Let us tell you why it **does** matter:

- Removing any sharp /undermined enamel.
- Removing irregularities provides better retention of the crown.
- Improve marginal fit of the crown.
- Reduce stress concentration.

Conclusion: Do **NOT** skip finishing if you want that A+ Prep!

Finishing Technique



How to check your work?

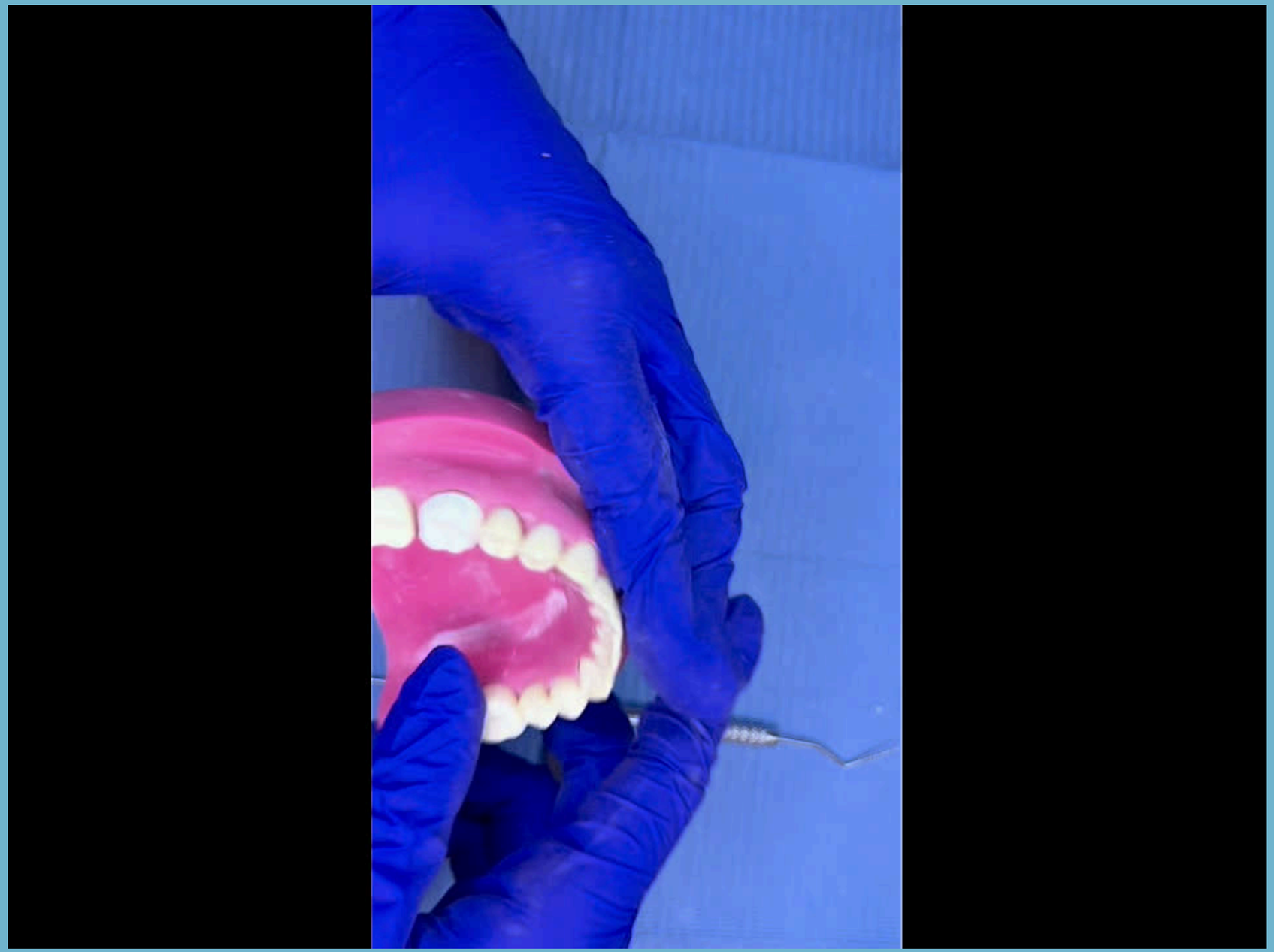
To check your work always come back to this checklist:



Check the AMOUNT of reduction done using index and probe	check that occlusal anatomy is preserved	check that there is no roughness or any sharp angles
Check that finish line is continous, all on the same level and has no irregularities	check that your taper is correct -no undercuts/over convergence	
check that there is no injury to adjacent tooth	check that you did the functional cusp bevel	
check that the contact is open properly	check that secondary plane is done properly	



How to check your work?




Congrats ! You have
now mastered Full
Veneer Crown Prep!

Contact Us!

If you have any further questions, stuck in any step or want to show us your progress, feel free to contact us completely free of charge!

Our talented instructor Dr. Salma is waiting for you!

Want to book a one on one session? Just give us a DM!

TikTok: @dentahelp

https://www.tiktok.com/@dentahelp?_r=1&_t=ZS-97oeLMuArIr

Whatsapp: +201105723303



Thank You

